

Code-Layout, Readability and Reusability

Date	7 July 2024
Team ID	XXXXXX
Project Name	Personalized Networking Assistant
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing and indentation is maintained throughout the code.	Yes	Python coding standards followed.
2	Proper File Structure	Files and folders are logically organized.	Yes	Separate files are created for frontend, backend, AI modules, and storage.
3	Meaningful Variable Names	Variables reflect their purpose clearly.	Yes	Descriptive names such as user profile, suggestions, and feedback are used.
4	Function / Method Names	Functions have descriptive names.	Yes	Functions like generate_message(), check_fact(), and save_feedback() describe their purpose.
5	Code Comments	Inline and block comments explain logic.	Yes	Comments are added for better understanding and maintenance.
6	Modular Design	Code is split into reusable functions/modules.	Yes	Backend, frontend, AI processing, and fact checking are separated.
7	No Redundant Code	Duplicate or unused code is removed.	Yes	Code is optimized and unnecessary repetition is avoided.
8	Error Handling	Exceptions and errors are handled properly.	Yes	Try-except blocks are used for file handling and API operations.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technolog	yWhere Reused	Reusability Level (High / Medium / Low)
1	FastAPI Backend	Python, FastAPI	API request handling	High

2	Streamlit Interface	Python, Streamlit	User interaction and result display	High
3	Theme Extraction Module	DistilBERT, Hugging Face	Extracting important topics from user input	Medium
4	Topic Generator Module	GPT-2, Python	Generating networking suggestions	Medium
5	Fact Checker Module	Wikipedia API	Verifying information	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Project files are organized into separate modules.
Readability	5	Clear naming conventions and comments are followed.
Reusability	4	Modules can be reused and extended for future features.
Documentation / Comments	4	Important functions and logic are documented.
Overall Score	5	Code is maintainable, modular, and scalable.